The following text is adapted from Jhumpa Lahiri's 2003 novel *The Namesake*. Gogol is an elementary school student in Massachusetts.

> In art class, his favorite hour of the week, he carves his name with paper clips into the bottoms of clay cups and bowls. He pastes uncooked pasta to cardboard, and leaves his signature in fat brush strokes below paintings.

©2003 by Jhumpa Lahiri

1 Mark for Review

As used in the text, what do the words "carves" and "leaves" both most nearly mean?

(A) Marks

(B) Passes

(C) Turns

(D) Follows

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HighEssence scoring service

The sloping tile roofs and picturesque façade of Mission San Carlos Borromeo de Carmelo in Carmel, California, are enduring symbols of the Spanish contribution to Californian architecture. Elements of this style have been reproduced throughout the state—the design of Santa Clara University in Santa Clara, for example, is considered to _____ the style of the Spanish missions.

2 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) renovate

(B) imitate

(C) neglect

(D) reject

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HighestScoringService

In a garden, _____ the spread of unwelcome weeds such as nutgrass can be difficult because weeds usually spread easily and quickly become unmanageable, making it hard to remove them completely.

3 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) retrieving

(B) saving

(C) halting

(D) pretending

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HighestScoringService

Certain features are almost always included in the designs of mosques, like the minaret (or tower), which is considered to be a _____ of mosque architecture. Even mosques that exhibit elements of multiple architectural styles, such as the Shah Jahan Mosque, which incorporates elements from the Mughal, Safavid, and Timurid styles, will also include several of these standard features.

4 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) hallmark

(B) rejection

(C) motivation

(D) duplicate

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Highest scoring service

Benjamin Prud'homme and colleagues have explored how convergent evolution—a phenomenon that occurs when the same trait evolves independently in two reproductively separate lineages—can result from a genetic mechanism shared by both lineages. Meanwhile, Patricia J. Wittkopp and colleagues have investigated how convergence occurs through different genetic mechanisms, but the relative prevalence of convergence through shared and different genetic processes is still poorly understood. This motivated biologists Delbert A. Green II and Cassandra G. Extavour to evaluate both types of convergence in a single study for their 2012 paper.

5 Mark for Review

Which choice best states the function of the underlined portion in the text as a whole?

(A) It provides examples of how a phenomenon was studied by scientists in the field before Green and Extavour's study.

(B) It gives a basic description of a phenomenon that is central to the discussion that follows.

(C) It clarifies a concept that the author implies was unclear in the studies mentioned in the text.

(D) It introduces a method of scientific analysis that is discussed in greater detail later in the text.

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Highest scoring service

In Ojibwe, an Indigenous language from the Great Lakes region of what are now the United States and Canada, *wàsa* means "far," whereas *wa-wàsa* means "far apart." This phenomenon, in which an element of a root word is repeated, sometimes with modification, within another word that is related to the root word, is called reduplication. In this case, the element "wà" in *wàsa* gets repeated in *wa-wàsa*. There are many examples of this type of reduplication in Ojibwe.

6 Mark for Review

Which choice best describes the overall structure of the text?

(A) It describes the relationship between Ojibwe and several other languages, raises a question about the nature of that relationship, and then answers that question.

(B) It identifies the most frequently occurring words in Ojibwe, explains why it is difficult to translate those words into English, and then provides examples of languages other than English into which those words can be translated.

(C) It presents some specific words in Ojibwe, describes the general linguistic phenomenon exemplified by those words, and then states that this phenomenon occurs frequently in Ojibwe.

(D) It explains the phenomenon of reduplication, discusses why reduplication has been controversial among scholars, and then argues that an analysis of Ojibwe could help resolve that controversy.

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HighestScore scoring service

Text 1

Uisdean Nicholson and his team have discovered evidence in seismic data of a 40-kilometer-wide subsurface crater beneath nearly a kilometer of water off the coast of West Africa that is consistent with a 400-meter-wide asteroid striking the seafloor. This structure, which the team named Nadir, exhibits all the telltale signs of a high-velocity impact crater: an elevated rim, a circular shape, a terraced floor, and a pronounced area of uplift at its center.

Text 2

Both carbonate dissolution and subsurface salt withdrawal can cause craterlike depressions without the need for a high-velocity impact. However, carbonate dissolution is very unlikely to have occurred in the vicinity of Nadir, and although subsurface salt withdrawal could have plausibly occurred in this area and would result in a depression with a terraced floor or a circular shape, it would not exhibit the area of central uplift seen at Nadir.

7 Mark for Review

Which choice best describes a difference between the approach of Text 1 and the approach of Text 2?

(A) Text 1 dispassionately describes Nicholson and colleagues' findings and conclusions, whereas Text 2 attempts to convey the researchers' excitement on discovering Nadir.

(B) Text 1 focuses on features Nadir lacks, whereas Text 2 indicates features it shares with other geological depressions.

(C) Text 1 discusses a single plausible cause of Nadir, whereas Text 2 evaluates two possible causes.

(D) Text 1 emphasizes the evidence supporting an asteroid impact as the cause of Nadir, whereas Text 2 argues against that explanation.

The following text is from Julia Alvarez's 2000 novel *In the Name of Salomé*. The narrator and her sister, daughters of a famous poet, are being tutored by Alejandro Román.

> Our tutor, Alejandro Román, brought his younger brother, Miguel, to class one day. By now I was eighteen and had learned everything Alejandro had to teach me, so I was glad for a new face. Miguel was an aspiring poet, and he had heard from his brother that the Ureña girls were none other than the daughters of Nicolás Ureña, and they were smart as clockwork. Miguel was hoping not only to meet us but to make the acquaintance of the poet himself at Mamá's house.

©2000 by Julia Alvarez

8 Mark for Review

Based on the text, why does Miguel accompany his brother to the sisters' house one day?

(A) Miguel has learned all his brother can teach him and now desires to be taught by the sisters.

(B) Miguel has not received formal instruction in poetry and wants to ask the sisters' famous father to be his mentor.

(C) Miguel wants to recite his poems to the sisters and inquire about their perspectives on his work.

(D) Miguel anticipates having the opportunity to be introduced to both the sisters and their father.

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Highest scoring service

Motivated to sell as many paintings as possible, Alfred Hair, an influential figure among the landscape artists known as the Florida Highwaymen, pioneered "fast painting," which in part involved working across multiple canvasses at once. That many of Hair's acolytes, including Isaac Knight, imitated the technique accounts in part for the impressionistic qualities that are now synonymous with the group's shared aesthetic. But not all Highwaymen fully embraced this approach; for instance, though Willie Reagan was also prolific, his paintings were executed with greater attention to detail.

9 Mark for Review

What does the text most strongly suggest about paintings by Knight?

(A) Because of the manner in which they were created, they likely have visual qualities that are regarded as more typical of Florida Highwaymen paintings than the qualities in works by Reagan are.

(B) Although it is evident that Knight adopted some of Hair's preferred techniques, Knight's works are less derivative of works by Hair than is typically acknowledged.

(C) The lack of precision with which they were executed suggests that they are inferior to works by either Hair or Reagan.

(D) Knight's reliance on the technique of fast painting likely accounts for his works being more aesthetically interesting than works by Reagan are.

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Highest scoring service

Vancouver has high pedestrian traffic, but other cities cannot increase their pedestrian traffic simply by replicating a single feature of Vancouver—e.g., its high number of pedestrian amenities—that is associated with walkability. As urbanist Mariela Alfonzo argues, many factors influence people's decision-making about whether to walk: some studies have shown the importance of demographic characteristics, others have shown the importance of average commuting distance, and so on, and it is clear that none of these factors in isolation fully explains pedestrian habits in a given city.

10 Mark for Review

Based on the text, the author would most likely agree with which statement about Vancouver's "high number of pedestrian amenities"?

(A) It may increase walkability in Vancouver but is known to reduce walkability in other cities.

(B) It is better understood as an effect of the high level of pedestrian traffic in Vancouver than as a cause of that pedestrian traffic.

(C) It affects walking decisions in Vancouver less than demographic characteristics and average commuting distance do.

(D) It should be understood as just one of several factors that influence pedestrian activity in Vancouver.

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Highest scoring service

Numbers of the 23 Non-native Tree Species Reported and the Insect and Fungus Threats to Them

Country	Trees	Fungi	Insects
Great Britain	18	290	120
Hungary	1	18	13
Switzerland	11	43	78

Elisabeth Pötzelsberger and colleagues gathered data on 23 non-native tree species grown in Europe. They analyzed reports from Great Britain, Switzerland, and Hungary about the number of these species grown in those countries as well as the numbers of insect and fungus species that damage those trees. The researchers concluded that Great Britain had a greater number of damaging fungus species than either of the other countries did.

11 Mark for Review

Which choice best describes data from the table that support Pötzelsberger and colleagues' conclusion?

(A) Hungary reported 18 damaging fungus species but only 13 damaging insect species.

(B) Great Britain reported 290 damaging fungus species, whereas Switzerland reported 78 damaging insect species.

(C) Great Britain reported 290 damaging fungus species, which is more than either Switzerland or Hungary reported.

(D) Switzerland and Hungary reported 11 and 1 damaging fungus species, respectively, which is far fewer than Great Britain reported.

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Highest scoring service

Poems is an 1895 collection of poetry by Frances E.W. Harper. In one of Harper's poems, the speaker criticizes activists who champion humanitarian causes in other countries while overlooking local concerns, saying, _____

12 Mark for Review

Which quotation from *Poems* most effectively illustrates the claim?

(A) "Men may tread down the poor and lowly, / May crush them in anger and hate, / But surely the mills of God's justice / Will grind out the grist of their fate." (from "An Appeal to My Countrywomen")

(B) "When ye plead for the wrecked and fallen, / The exile from far-distant shores, / Remember that men are still wasting / Life's crimson around your own doors." (from "An Appeal to My Countrywomen")

(C) "God bless our native land, / Land of the newly free, / Oh may she ever stand / For truth and liberty." (from "God Bless Our Native Land")

(D) "Let me make the songs for the people, / Songs for the old and young; / Songs to stir like a battle-cry / Wherever they are sung." (from "Songs for the People")

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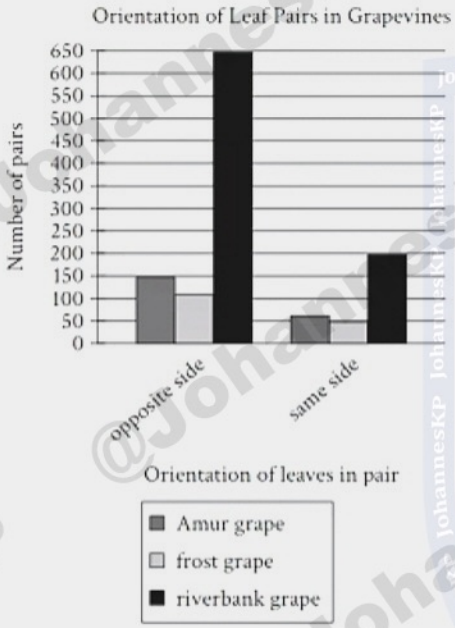
Highest scoring service



Many plants have leaves that are larger on one side of their long central axis than the other, a phenomenon known as asymmetrical orientation. University of California, Berkeley biologist Ciera Martinez and colleagues examined several species of grapevines and closely related plants, which have leaves that grow in pairs, to see if both leaves in a pair tend to be oriented toward the same side (that is, have more tissue on the same side) or not. They found that opposite-side orientation was much more common than same-side orientation; in the Amur grape, for example, approximately _____

13 Mark for Review

Which choice most effectively uses data from the graph to complete the example?

(A) 150 leaf pairs show opposite-side orientation, whereas approximately 60 pairs show same-side orientation.

(B) 110 leaf pairs show opposite-side orientation, whereas approximately 45 pairs show same-side orientation.

(C) 650 leaf pairs show opposite-side orientation, whereas no pairs show same-side orientation.

(D) 110 leaf pairs show opposite-side orientation, whereas no pairs show same-side orientation.

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Mooseberry (*Viburnum edule*) plants are native to Alaska, where harsh conditions have historically impeded potential invasive species. As the boreal climate has warmed in recent decades, however, common knotgrass (*Polygonum aviculare*) plants have established themselves in Alaska. It has been suggested that warming-induced delays in the onset of subfreezing temperatures in autumn can benefit invasives more than native species; to evaluate this possibility, biologists Christa Mulder and Katie Spellman tracked *V. edule* and *P. aviculare*, along with other native and invasive species, over several years, concluding that invasives are advantaged by delays in subfreezing temperature onset in Alaska.

14 Mark for Review

Which finding, if true, would most directly support Mulder and Spellman's conclusion?

(A) Although *V. edule* and *P. aviculare* both tended to produce leaves later into autumn in years with late subfreezing temperature onset, the extension was much greater for *P. aviculare* than for *V. edule*.

(B) Although significant interannual variations in subfreezing temperature onset were observed during the study, neither *P. aviculare* nor *V. edule* showed any significant interannual variation in the cessation of leaf production.

(C) Although *V. edule* and *P. aviculare* tended to stop producing leaves at about the same time in years with historically typical temperature patterns, *P. aviculare* stopped producing leaves sooner than *V. edule* did in years with late subfreezing temperature onset.

(D) Although *V. edule* and *P. aviculare* both tended to produce more leaves overall in years with late subfreezing temperature onset than they did in years with historically typical temperature patterns, the years with late subfreezing temperature onset also had early growing season onset in spring.

Northeastern Kansas's Jefferson County is among the most rural counties in the United States: the US Census Bureau classified it as 98.8% rural in 2010. Researchers studying populations of counties like Jefferson often struggle to recruit and retain participants. Melissa Valerio and colleagues tested whether a method called snowball sampling could improve recruitment and retention. Working in two rural counties, the researchers identified a small number of people who had the characteristics desired for a proposed study and asked them to recruit additional participants from their social networks. Valerio and colleagues found that participants recruited via snowball sampling showed a much higher retention rate than did people recruited by strangers, suggesting that _____

15 Mark for Review

Which choice most logically completes the text?

(A) being recruited to participate in a study by someone with whom one is socially connected may impart a feeling of obligation to persist with participation in the study.

(B) people with relatively small social networks are inherently less likely to be recruited to participate in a study via snowball sampling than are people with relatively large social networks.

(C) snowball sampling is more likely to improve retention rates among rural participants than among nonrural participants.

(D) social networks can become large enough that two people can share a network but nevertheless regard each other as strangers.

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Highest scoring service

Miniature Self-Portrait (1556) is one of dozens of remaining works by Sofonisba Anguissola. Currently, her painting _____ at the Museum of Fine Arts in Boston, where visitors can view the Italian Renaissance artist's work on display.

16 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) will have hung

(B) hangs

(C) hung

(D) was hanging

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Highest scoring service

The human foot contains a variety of skeletal muscles, each playing an important role in the foot's healthy movement. The extensor digitorum brevis, which is attached to the calcaneus, _____ responsibility for extending the middle toes.

17 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) bearing

(B) to bear

(C) bears

(D) having borne

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Highest scoring service

Louisiana resident Oscar James Dunn, one of the nearly two thousand African Americans elected to public office during the decade that followed the Civil War, _____ his term as lieutenant governor in 1868.

18 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) to begin

(B) having begun

(C) beginning

(D) began

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Highest scoring service

The giant sequoia (*Sequoiadendron giganteum*) known as D-23, located in the United States, was one of the oldest known trees in the world, at 3,075 years old. With over three millennia of climate data in its tree _____ single tree like this, claims dendrochronologist Valerie Trouet, can tell the history of the world.

19 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) rings, A
(B) rings; a
(C) rings and a
(D) rings, a

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Highest scoring service

After finding information about Blanche Kelso Bruce, who represented Mississippi in the United States Senate, the student discovered biographical sketches of two other Black Americans who served in _____ Ralph Harold Metcalfe of Illinois and Alton R. Waldon Jr. of New York.

20 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) Congress:

(B) Congress;

(C) Congress.

(D) Congress

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Highest scoring service

John Thomson employed the pseudonym "Gracchus"—the name of an ancient Roman politician—in political essays he wrote in 1795, a choice that accomplished far more than simply concealing his authorship. _____ it wasn't an arbitrary pen name but rather a complex rhetorical strategy through which Thomson aligned his political views with the venerated republican ideals of the ancient world, thereby bolstering the authority of his writing.

21 Mark for Review

Which choice completes the text with the most logical transition?

(A) Indeed,

(B) Conversely,

(C) In addition,

(D) However,

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HighestScoring Service

In Annie Dillard's *Pilgrim at Tinker Creek*—where, early on, the author marvels at a single goldfish's delicate fins but later winces when imagining a horde of goldfish laying and eating their own eggs—Dillard struggles to reconcile the complicated juxtapositions of the natural world. _____ nature's mesmerizing intricacy and pitiless harshness prove inextricably linked for Dillard, like "two branches of the same creek."

22 Mark for Review

Which choice completes the text with the most logical transition?

(A) Ultimately,

(B) To that end,

(C) Hence,

(D) Moreover,

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Highest scoring service

While researching a topic, a student has taken the following notes:

- The Galápagos penguin is a species of bird that can be found on the Galápagos Island of Floreana.
- It has an average weight of 2.5 kilograms.
- The grey warbler finch is a species of bird that can be found on the Galápagos Island of Floreana.
- It has an average weight of 9 grams.

23 Mark for Review

The student wants to emphasize a similarity between the Galápagos penguin and the grey warbler finch. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) On average, the Galápagos penguin weighs 2.5 kilograms, while the grey warbler finch weighs 9 grams.

(B) The grey warbler finch, which weighs 9 grams on average, can be found on the Galápagos Island of Floreana.

(C) The grey warbler finch and the Galápagos penguin can both be found on the Galápagos Island of Floreana.

(D) The Galápagos Island of Floreana is home to several bird species, one of which is the Galápagos penguin.

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Highest scoring service

While researching a topic, a student has taken the following notes:

- The Mohs scale of mineral hardness is a ten-point scale that orders minerals by hardness based on their ability to scratch other minerals.
- Minerals with larger numbers are harder than minerals with smaller numbers and can leave visible scratches on them.
- Minerals with smaller numbers are softer than minerals with larger numbers and cannot leave visible scratches on them.
- The mineral fluorite has a Mohs scale number of 4.
- The mineral orthoclase has a Mohs scale number of 6.
- The mineral corundum has a Mohs scale number of 9.

24 Mark for Review

The student wants to emphasize orthoclase's Mohs scale number. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) In the Mohs scale of mineral hardness, corundum (9) is ranked higher than fluorite (4).

(B) Orthoclase has a Mohs scale number of 6, which means that it is harder than fluorite (4) but softer than corundum (9).

(C) Orthoclase, fluorite, and corundum can be ordered by their ability to leave visible scratches on other minerals.

(D) Corundum, which has a Mohs scale number of 9, can scratch both fluorite and orthoclase.

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Highest scoring service

While researching a topic, a student has taken the following notes:

- Zora Neale Hurston was an acclaimed writer.
- Her first published work of fiction was a short story.
- It was called "John Redding Goes to Sea."
- It first appeared in *Stylus* in 1921.

25 Mark for Review

The student wants to identify the title of Zora Neale Hurston's first published short story. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) Zora Neale Hurston's first published work of fiction appeared in 1921.

(B) Zora Neale Hurston's first published short story was called "John Redding Goes to Sea."

(C) In 1921, a short story by Zora Neale Hurston appeared in *Stylus*.

(D) Acclaimed writer Zora Neale Hurston's first published work of fiction was a short story.

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Highest scoring service

While researching a topic, a student has taken the following notes:

- Body positions are a fundamental aspect of dance.
- In Vaganova method ballet, there is a position for the dancer's arms called first position.
- In the position, both arms are rounded to roughly align the fingers with the navel.
- In Royal Academy of Dance ballet, there is a position for the dancer's arms called demi-seconde position.
- In this position, both arms are out to the sides but angled downward.

26 Mark for Review

The student wants to emphasize a similarity between the two positions. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) First position in Vaganova method ballet and demi-seconde position in Royal Academy of Dance ballet are both positions for the dancer's arms.

(B) Both first position and demi-seconde position are positions in Royal Academy of Dance ballet.

(C) In first position, both arms are rounded to roughly align the fingers with the navel, in contrast to demi-seconde position, where both arms are out to the sides but angled downward.

(D) In Vaganova method ballet, there are a number of positions, including first position.

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Highest scoring service

While researching a topic, a student has taken the following notes:

- *Save Me* is a 2018 sculpture by American artist Olivia Steele.
- It is a pink neon sign that spells out the title phrase.
- Neon signs became popular with advertisers in the 1920s and began to decline in popularity in the 1960s.
- Neon signs became popular with artists in the 1960s and continue to be popular today.
- Writer Sarah Archer says, "[Neon signs] must be made by hand because there's no cost-effective way to mass-produce them."

27 Mark for Review

The student wants to connect the sculpture to the history of neon in art. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) Neon signs like *Save Me* are popular with advertisers and artists alike, though they must, as Archer notes, be made by hand, as they cannot be easily mass-produced.

(B) With *Save Me*, Steele joins the decades-long lineage of artists who have helped make neon a popular artistic medium since the 1960s.

(C) Continuing from the 1920s to today, neon lighting has been a popular material, both as an advertising tool and an artistic medium.

(D) The pink neon sign *Save Me* nods to the rich history of a material whose artistic popularity has long been in decline.

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Highest scoring service

No reading and writing module 2

Because the reading and writing module 2 of the US test paper B and the reading and writing module 2 of the US test paper A and the reading and writing module 2 of the US test paper C are mostly repeated

The following reading and writing sections will still be Module 1 it is from reading and writing module 1 of the US test paper C

@JohannesKP

The following text is adapted from Yone Noguchi's 1914 memoir *The Story of Yone Noguchi*. Noguchi is returning home after eleven years abroad.

I reached Tsushima, my native town, at evening.

I frightened my old father at the station, who was actually trying to find me among some other people. There is no wonder that he could not recognise me; I must have changed a great deal.

1 Mark for Review

As used in the text, what does the word "find" most nearly mean?

(A) Reveal

(B) Persuade

(C) Judge

(D) Locate

Lessons from previous declines in biodiversity can help shape policy recommendations aimed at preserving species at risk. The factors that contributed to the extinction of the shrub-ox around 9500 BCE may also be _____ endangered species today.

2 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) concealed from

(B) applicable to

(C) integrated with

(D) dependent on

@JohannesKP Highest scoring service

In a 2018 article celebrating films depicting the Black experience, critics for the *New York Times* ___ Ossie Davis's 1970 film *Cotton Comes to Harlem* and Cheryl Dunye's 1996 film *The Watermelon Woman*, praising the former as "especially pointed" and the latter as "heartbreaking and inspiring."

3 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) inspired

(B) satirized

(C) overlooked

(D) commended

@JohannesKP
Highest scoring service

In 1879, in an effort to determine the longevity of certain seeds left in soil, botanist William Beal buried 20 glass bottles filled with sand and seeds from a variety of plants, including *Verbascum*, *Lepidium virginicum*, and *Stellaria media*. Since then, some of the bottles have been retrieved in the periodic checks of seed viability researchers are conducting. After 60 years had elapsed, most of the seeds in those bottles stopped germinating, but *Verbascum* seeds have remained viable, including those from a bottle dug up in 2021 by Margaret Fleming, Lars Brudvig, and colleagues.

4 Mark for Review

Which choice best states the main purpose of the text?

(A) To discuss an ongoing experiment

(B) To explain how to get rid of certain plants

(C) To compare two experiments addressing the same questions

(D) To describe difficulties associated with preserving seeds

Mexican textile artist Victoria Villasana weaves stories of triumph, using her unique method of applying colorful yarn to photographs of people. In some works, Villasana focuses on celebrating cultural icons who are people of color, as she does in her depiction of musician and composer Prince. However, in other works, Villasana honors ordinary people, as she does in her captivating portrayal of a young girl sitting on a sidewalk. Villasana sees both of these approaches as ways of depicting the power and interconnectedness of all people.

5 Mark for Review

Which choice best describes the overall structure of the text?

(A) It introduces Villasana's cultural background, describes how this background influences her art, and then outlines some of her ideas for future projects.

(B) It compares Villasana to other contemporary artists, indicates how two of her works are similar, and then emphasizes Villasana's enthusiasm for artistic collaboration.

(C) It explains how yarn has historically been used in art, details how Villasana uses yarn in her artworks, and then comments on the popularity of her work.

(D) It provides details about Villasana's artworks, discusses specific examples of her work, and relates them to one of her artistic goals.

The following text is from Sara Teasdale's 1922 poem "Two Songs for Solitude."

My heart has grown rich with the passing of years,
I have less need now than when I was young
To share myself with every comer,
Or shape my thoughts into words with my tongue.

6 Mark for Review

Which choice best describes the function of the underlined portion in the text as a whole?

(A) It explains the speaker's views about proper conduct toward new acquaintances.

(B) It provides examples of behaviors the speaker is less inclined toward than she once was.

(C) It presents an episode from the speaker's youth that she remembers fondly.

(D) It emphasizes the social habits the speaker has developed over her lifetime.

@JohannesKP

Text 1

In 2015 Rasht, Iran, was named a City of Gastronomy by UNESCO in recognition of the city's unique food culture. The honor is well known among residents of the city as well as tourists who visit. It's obvious that the selection of Rasht by UNESCO has brought awareness to local recipes, cooking practices, and chefs and has provided a boost to the city's tourism industry.

Text 2

Many people in Rasht's restaurant industry hoped that food tourism would increase after the city was chosen as a City of Gastronomy in 2015. However, as researcher Eerang Park and colleagues argue, cities must still create effective marketing strategies to benefit fully from being named a City of Gastronomy. Without an intentional effort to promote the city's food scene, many current and potential visitors to Rasht may not even be aware that it's home to uniquely delicious food.

7 Mark for Review

Based on the texts, both authors would most likely agree with which statement?

(A) The benefits of Rasht having been named a City of Gastronomy extend well beyond increased tourism.

(B) Increased tourism isn't guaranteed after a city has been named a City of Gastronomy.

(C) A city's food scene can benefit from the city being named a City of Gastronomy.

(D) A significant number of visitors to Rasht may not know that it was named a City of Gastronomy.

Like many other bird species that live only on the Hawaiian archipelago, the ʻakekeʻe has adapted to life in a well-defined habitat, resulting in highly specialized physical and behavioral characteristics that aid the species in survival. However, because the ʻakekeʻe is highly specialized, it is especially vulnerable to environmental changes that can disrupt the delicately balanced ecosystem in which it lives.

8 Mark for Review

Which choice best states the main idea of the text?

(A) Hawaiian birds display a unique range of physical and behavioral characteristics and as a result can only live in habitats unique to the Hawaiian archipelago.

(B) The ʻakekeʻe is an example of a species unique to the Hawaiian archipelago that is highly specialized and therefore particularly susceptible to habitat disturbances.

(C) The ʻakekeʻe is a species of bird that is related to many other Hawaiian birds but does not share a habitat with any of them.

(D) The ʻakekeʻe is an example of a highly specialized bird species found only on the Hawaiian archipelago and is related to several other highly specialized bird species found there.

@JohannesKP
Highest scoring service

In 2004, Andrew Rogers created *Ancient Language*. In this work, Rogers created an 80-meter-long geoglyph out of rocks in the Atacama Desert. Rogers's work is part of the land art movement, in which artists reject the confinement and commercialization of art galleries. These artists usually install large-scale pieces outdoors, often in surprising locations.

9 Mark for Review

Based on the text, what can be inferred about *Ancient Language*?

(A) Its location may be considered surprising.

(B) It marked a shift in the popularity of the land art movement.

(C) It may have had more commercial success in a different outdoor location.

(D) It was created early in Rogers's career.

@JohannesKP

Home Video Game Systems of the 1970s and 1980s

System	Manufacturer	Approximate number of units sold worldwide
ColecoVision	Coleco	2,000,000
Game & Watch	Nintendo	18,600,000
Intellivision	Mattel	3,000,000
Apple II	Apple Inc.	4,487,000

A student is researching the ColecoVision and other important gaming systems that were part of the global rise of the home video game industry during the 1970s and 1980s. The student is surprised to find that the ColecoVision sold relatively few units worldwide, with only about _____

10 Mark for Review

Which choice most effectively uses data from the table to complete the statement?

(A) 3,000,000 units sold compared to the approximately 4,487,000 units sold of the Game & Watch.

(B) 2,000,000 units sold compared to the approximately 18,600,000 units sold of the Game & Watch.

(C) 2,000,000 units sold compared to the approximately 18,600,000 units sold of the Intellivision.

(D) 3,000,000 units sold compared to the approximately 4,487,000 units sold of the Apple II.

A student is studying how woma pythons, eastern brown snakes, and rough-scaled death adders react to sound. The student knew that many snakes display movement of more than 20 cm in response to sound traveling through the ground, but assumed that snakes do not detect sound traveling through the air until reading a study by Christina Zdenek, Timothy Staples, and their team.

11 Mark for Review

Which finding from Christina Zdenek, Timothy Staples, and their team's study, if true, would most directly challenge the underlined assumption?

(A) Neither eastern brown snakes nor rough-scaled death adders display movement of more than 20 cm in response to airborne sound.

(B) Woma pythons have better overall hearing than do rough-scaled death adders.

(C) Both woma pythons and eastern brown snakes react only to sounds traveling through the ground.

(D) Woma pythons, eastern brown snakes, and rough-scaled death adders display movement of more than 20 cm in response to both ground and airborne sound waves.

"A Pair of Silk Stockings" is an 1897 short story written by Kate Chopin. In the story, Mrs. Sommers becomes engrossed in the decision of how she should spend a recently obtained sum of money: _____

12
Mark for Review

Which quotation from "A Pair of Silk Stockings" most effectively illustrates the claim?

(A) "The question of investment was one that occupied her greatly. For a day or two she walked about apparently in a dreamy state, but really absorbed in speculation and calculation."

(B) "She had learned to clutch a piece of goods and hold it and stick to it with persistence and determination till her turn came to be served, no matter when it came."

(C) "A young girl who stood behind the counter asked her if she wished to examine their line of silk hosiery. She smiled, just as if she had been asked to inspect a tiara of diamonds with the ultimate view of purchasing it."

(D) "It seemed to her a very large amount of money, and the way in which it stuffed and bulged her worn old *porte-monnaie* [small purse] gave her a feeling of importance such as she had not enjoyed for years."

@JohannesKP

Australia	8.8	-2.3	6.8	7.1
Guatemala	13.5	16.7	13.8	20.1
Jordan	-5.3	42.1	-5.5	36.7
Morocco	12.7	42.8	19.6	4.9
Panama	-6.8	7.3	10.0	11.0

Data are approximate.

A 2022 US Department of Agriculture report by Kayode Ajewole et al. calculated average annual growth rates of agricultural exports from countries over the five years before and the five years following the creation of a free trade agreement (FTA) with the US. The table shows data for five countries in the study. (Post-FTA calculations included some anticipatory effects preceding the agreements' official start.) Ajewole et al. note that an increase in the rate of exports to the US in the post-FTA period does not necessarily indicate that a country produced more goods for export as a result of the FTA. Rather, FTAs sometimes incentivize countries to redirect existing trade from nonmember countries to FTA partners, as is most likely the case with _____

13 Mark for Review

Which choice most effectively uses data from the table to complete the statement?

(A) Jordan, because the post-FTA period coincided with increasing rates of both its agricultural exports to the US and its total agricultural exports to countries not participating in the FTA.

(B) Australia, because its rate of agricultural exports to the US and its rate of total agricultural exports both decreased in the post-FTA period relative to the pre-FTA period.

(C) Morocco, because its rate of agricultural exports to the US increased in the post-FTA period relative to the pre-FTA period, while its rate of total agricultural exports decreased during the same period.

(D) Panama, because the post-FTA period saw a decrease in its rate of agricultural exports to the US but not in its rate of total agricultural exports.

Studies conducted in individual Western societies have demonstrated associations between personality traits and five musical factors (mellowness, unpretentiousness, sophistication, intensity, and contemporariness) underlying individual preferences for styles of Western music. To investigate these associations across cultures, David M. Greenberg et al. collected music-preference assessments for Western genres and self-reported personality traits from participants in fifty-three countries across six continents. The study confirmed that the five-factor framework accurately captured participants' tastes in Western music, and, moreover, the study found similar correlations between patterns of these factors and of personality traits, suggesting that _____

14
Mark for Review

Which choice most logically completes the text?

(A) the five-factor framework can likely be used to predict preferences for non-Western music styles based on personality traits even if the characteristics of those styles substantially differ from characteristics of Western music styles.

(B) the strength of the relationship between personality traits and musical preferences varies less across cultures than researchers had previously assumed.

(C) people with a relatively high degree of familiarity with Western music styles are likely to express stronger preferences for those styles than people with a relatively low degree of familiarity with those styles are.

(D) across cultural contexts, people who share similar profiles of personality traits tend to prefer listening to similar types of

The oceanic whitetip shark and the Pacific halibut are ectothermic (cold-blooded) fish, whereas the shortfin mako shark and the yellowfin tuna are regional endotherms—they retain metabolic heat resulting in body temperatures above the ambient water temperature. Modeling the effect of falling ambient temperatures on ectotherms indicated to researchers Haley R. Dolton and colleagues that ectotherms' body temperatures inexorably decrease toward the ambient temperature. Data from wild basking sharks show that even though their body temperature consistently remains 1.0 to 1.5°C above ambient, it also declines as the ambient temperature falls. This suggests that the body temperatures of the _____

15 Mark for Review

Which choice most logically completes the text?

(A) yellowfin tuna and the shortfin mako shark are likely to be more variable than the body temperature of the basking shark.

(B) Pacific halibut and the basking shark are at least partly determined by the ambient water temperature.

(C) basking shark and the shortfin mako shark remain fixed even as ambient water temperature changes, while the body temperature of the Pacific halibut changes with variations in the ambient water temperature.

(D) basking shark and the yellowfin tuna remain fixed even if the ambient water temperature changes, while the body temperature of the oceanic whitetip shark varies even if the ambient temperature is stable.

Nanoengineers have developed synthesized cerium oxide nanoparticles that are used to improve treatments for certain conditions related to oxidative stress. As nanotechnology evolves, medical technology _____ to evolve as well.

16 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) had likely been continuing

(B) will likely continue

(C) would likely have continued

(D) had likely continued

@JohannesKP
Highest Scoring service

Mr. T, Rasta, and Makhulu were special animals to big cat enthusiasts. Between 2006 and _____ lions prowled the Greater Kruger National Park game reserve in South Africa, forming part of a dominant coalition—the term for a group of male lions—known as the Mapogo.

17 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) 2012 when these

(B) 2012. These

(C) 2012, because these

(D) 2012, these

@JohannesKP
Highest scoring service

Navajo artist Sally Black's mastery of traditional weaving techniques is apparent in her expertly crafted baskets, which typically are woven from sumac (a textured and flexible material) and _____ bold colors.

18 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) feature

(B) had featured

(C) were featuring

(D) featured

@JohannesKP Highest scoring service

A 2014 study led by _____ studied the impact of fertilizers containing nitrogen on grassland arthropod populations. Another study, led by Malie Lessard-Therrien in 2018, looked at fertilizers containing nitrogen and two other macronutrients: phosphorus and potassium.

19 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) researcher, Mark A. Lee,

(B) researcher Mark A. Lee,

(C) researcher Mark A. Lee

(D) researcher, Mark A. Lee

@JohannesKP
Highest scoring service

Most conifers (plants that grow cones, such as pines, firs, and spruces) are evergreen, meaning that they keep live foliage year-round. The tamarack tree (*Larix laricina*), however, is a deciduous conifer, meaning that it regularly _____ all its needles.

20 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) shed

(B) have shed

(C) sheds

(D) shedding

@JohannesKP
Highest scoring service

Among the many ancient Egyptian units of measurement, the units *sa* and *shesep* were used to measure area and _____ neither unit is commonly used today.

21 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) length; respectively,

(B) length respectively

(C) length, respectively;

(D) length, respectively,

@JohannesKP Highest scoring service

Textile artist Amber Joy Greenidge-Sabral wanted to avoid using harsh synthetic dyes that could be damaging to the environment. _____ she began experimenting with more environmentally friendly natural dye methods, using organic materials like Texas paintbrush and madder root for their natural brown and red colors.

22 Mark for Review

Which choice completes the text with the most logical transition?

(A) With this in mind,

(B) To sum up,

(C) Nevertheless,

(D) In actuality,

@JohannesKP

While researching a topic, a student has taken the following notes:

- In a 2021 study, researchers showed participants an unaltered image of a popular character or logo alongside two slightly altered versions.
- Fred Flintstone is a cartoon man in a spotted tunic.
- In the first alteration, the tunic had a striped pattern.
- In the second alteration, the tunic had no pattern.
- Participants were asked to identify the correct version.
- 86.9% of participants selected the unaltered image, and 13.1% selected an altered version.

23 Mark for Review

The student wants to emphasize how many participants selected an altered version of Fred Flintstone. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) When participants were asked to identify the correct version of Fred Flintstone, 86.9% selected the unaltered image.

(B) A 2021 study showed participants an unaltered image of Fred Flintstone, who is a cartoon man in a spotted tunic, alongside two altered versions.

(C) Participants were asked to identify the correct version of Fred Flintstone, who is a cartoon man in a spotted tunic.

(D) When shown three versions of Fred Flintstone and asked to select the correct one, 13.1% of participants chose an altered version.

While researching a topic, a student has taken the following notes:

- The human tongue contains taste receptors for a rich, savory flavor called umami.
- Umami is triggered by the compounds in a variety of foods, including tuna and garlic.
- Participants in a study tasted a sample of North Pacific konbu, a type of brown seaweed.
- They rated its umami intensity as moderate.
- The participants tasted a sample of arame, another type of brown seaweed.
- They rated its umami intensity as high.

24 Mark for Review

The student wants to emphasize a difference between the two seaweeds. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) Although North Pacific konbu and arame are types of brown seaweed, the latter's umami flavor is more intense.

(B) Some types of brown seaweed, like North Pacific konbu and arame, trigger umami flavor in human taste buds.

(C) While North Pacific konbu and arame contain umami flavor, umami can also be found in tuna and garlic.

(D) North Pacific konbu is a type of brown seaweed, but so is arame.

While researching a topic, a student has taken the following notes:

- *Hadramphus tuberculatus* is an insect species.
- It was believed to be extinct until a living specimen was identified in New Zealand in 2004.
- *Catagonus wagneri* is a mammal species.
- It was believed to be extinct until a living specimen was identified in Argentina in 1974.
- They are considered Lazarus species.
- "Lazarus species" is a term for living species of organisms that were once believed to be extinct.

25 Mark for Review

The student wants to specify where *Hadramphus tuberculatus* was identified. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) In 1974, a living specimen of *Catagonus wagneri* was found in Argentina.

(B) A living specimen of *Hadramphus tuberculatus*, once believed to be extinct, was identified in 2004.

(C) Previously believed to be extinct, a living specimen of *Hadramphus tuberculatus* was identified in New Zealand.

(D) Examples of Lazarus species can be found in *Hadramphus tuberculatus* as well as *Catagonus wagneri*.

- Herpetology is primarily the study of amphibians and reptiles.
- Mary Cynthia Dickerson was an American herpetologist born in 1866.
- She is known for her research on frogs.
- Biochemistry is primarily the study of the chemical makeup and activity of organisms.
- Muriel Wheldale Onslow was a British biochemist born in 1880.
- She is known for her research on the basis of petal color in dragon flowers.

26 Mark for Review

Which choice most effectively uses information from the given sentences to emphasize a similarity between Dickerson and Onslow?

(A) Mary Cynthia Dickerson conducted research on frogs; Muriel Wheldale Onslow, by contrast, is known for her research on the basis of petal color in dragon flowers.

(B) Herpetology and biochemistry are just two examples of the myriad fields in which scientists can specialize.

(C) Like herpetologist Mary Cynthia Dickerson, biochemist Muriel Wheldale Onslow dedicated her scientific career to the study of living organisms.

(D) The herpetologist Mary Cynthia Dickerson was born in 1866, but biochemist Muriel Wheldale Onslow was born later, in 1880.

- Generally, an object will heat up when twisted.
- The twisting of an object is known as torsion.
- A 2019 study led by researchers Zunfeng Liu and Ray Baughman tested the torsional heating of various fibers.
- When dimethicone fiber was twisted, its average surface temperature increased by 1.8°C.
- When natural rubber fiber was twisted, its average surface temperature increased by 15.3°C.

27 Mark for Review

Which choice most effectively uses information from the given sentences to explain the concept of torsional heating?

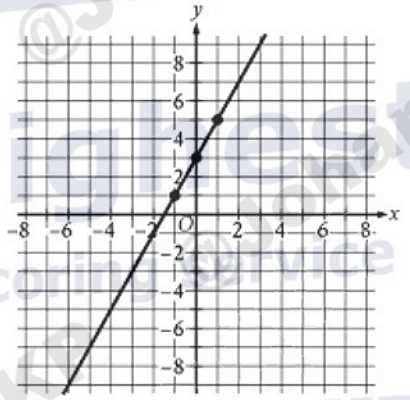
(A) Researchers determined that when both fibers were twisted, the surface temperature of natural rubber increased more than that of dimethicone.

(B) Torsional heating is the process by which an object increases in temperature when it is twisted.

(C) An object's average surface temperature will generally change due to torsion.

(D) In 2019, researchers studied the effect of torsional heating on various fibers, like dimethicone and natural rubber.

1 Mark for Review



The graph shows the linear relationship between x and y. Which table gives three values of x and their corresponding values of y for this relationship?

(A)

x	y
-1	1
0	0
1	5

(B)

x	y
-1	1
0	3

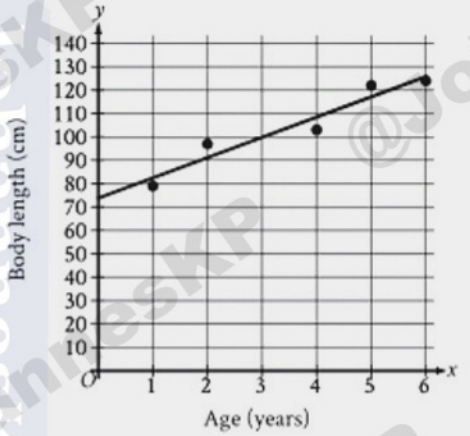
Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	31/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 0.666 0.667	0.66 .66 0.67 .67
$-\frac{1}{3}$	-1/3 -.3333 -0.333	-.33 -0.33

2 Mark for Review



The scatterplot shows 5 measurements of the body length, in centimeters (cm), of a New Zealand fur seal from an age of 1 year to 6 years old. A line of best fit is also shown. For a New Zealand fur seal at an age of 3 years old, what is the body length predicted by the line of best fit, to the nearest 10 cm?

Answer Preview:

Highest scoring service
@JohannesKP

3 Mark for Review

If $6x + 36 = 54$, what is the value of $x + 6$?

(A) 6

(B) 9

(C) 36

(D) 42

@JohannesKP
Highest scoring service

4 Mark for Review

For a certain classification, the positive mass m, in grams, of an object must be less than 510 grams. Which inequality represents this situation?

(A) $m > 510$

(B) $m < 0$

(C) $-510 < m < 0$

(D) $0 < m < 510$

@JohannesKP Highest scoring service

5 Mark for Review

The perimeter of an obtuse triangle is 27 meters. The lengths of two sides of this triangle are 5 meters and 9 meters. What is the length, in meters, of the third side of this triangle?

(A) 13

(B) 14

(C) 18

(D) 22

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Highest scoring service

6 Mark for Review

Task	Time (minutes)
A	8
B	5
C	12
D	10
E	10

The table shows the amount of time it took a participant in a study to complete each of 5 tasks. What was the mean time, in minutes, for this participant to complete a task?

(A) 7

(B) 9

(C) 10

(D) 12

@JohannesKP
Highest scoring service

7 Mark for Review

A group of 40 employees selected at random from all the employees at a company were surveyed about the number of books they read last year. From the data collected, it was estimated that the mean number of books employees at the company read last year is 6.8, with an associated margin of error of 1.1. Which of the following is the most appropriate conclusion?

(A) It is plausible that the actual mean number of books employees at the company read last year is less than 5.7.

(B) It is plausible that the actual mean number of books employees at the company read last year is between 5.7 and 7.9.

(C) It is plausible that every employee at the company read between 6.8 and 10.1 books last year.

(D) It is plausible that the actual mean number of books employees at the company read last year is greater than 7.9.

8 Mark for Review

Which expression is equivalent to $(5x^3 - 3x + 8) - (9x^6 + 5x - 3)$?

(A) $-4x^9 - 8x - 11$

(B) $-9x^6 + 5x^3 - 8x + 11$

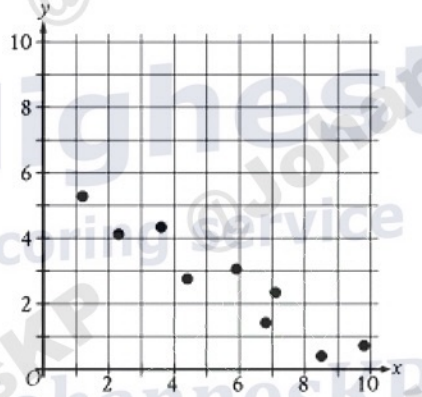
(C) $-9x^6 + 5x^3 + 2x + 5$

(D) $-4x^9 - 2x + 5$

@JohannesKP

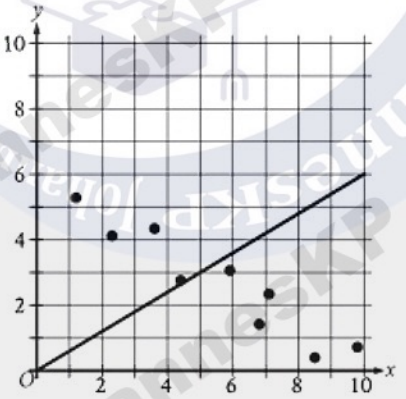
9 Mark for Review

The scatterplot shows the relationship between two variables, x and y.



Which of the following graphs shows a line of best fit that is most appropriate for the data shown in the scatterplot?

A



Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
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- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

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$\frac{2}{3}$	2/3 .6666 .6667 0.666 0.667	0.66 .66 0.67 .67
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10 Mark for Review

The function $f(x) = 180(x - 2)$ gives the sum of the interior angles, in degrees, for a polygon with x sides. What is the sum of the interior angles, in degrees, for a polygon with 35 sides?

Answer Preview:

@JohannesKP
Highest scoring service

11 Mark for Review

x	$h(x)$
0	17
1	18
2	20

For the function h, the table gives three values of x and their corresponding values of $h(x)$. Which equation could define h?

(A) $h(x) = 2^x + 16$

(B) $h(x) = 2^x + 15$

(C) $h(x) = 2^x + 2$

(D) $h(x) = 2^x$

@JohannesKP
Highest scoring service

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
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12 Mark for Review

$$f(x) = 3(x-1)(x-8)(x-11)$$

If the given function f is graphed in the xy-plane, where $y = f(x)$, what is the x-coordinate of an x-intercept of the graph?

Answer Preview:

@JohannesKP
Highest scoring service

13 Mark for Review

$$x + 4y = 19$$
$$2x + y = 4$$

The solution to the given system of equations is (x, y). What is the value of $3x + 5y$?

(A) 12

(B) 15

(C) 23

(D) 38

@JohannesKP
Highest scoring service

14 Mark for Review

In triangle DEF, the measure of angle D is 47° and the measure of angle E is 97°. In triangle RST, the measure of angle R is 47° and the measure of angle S is 97°. Which of the following additional pieces of information is needed to determine whether triangle DEF is similar to triangle RST?

A The measure of angle F

B The measure of angle T

C The measure of angle F and the measure of angle T

D No additional information is needed.

@JohannesKP

15 Mark for Review

$$x^2 + y = 21$$
$$-x + y = 1$$

The solutions to the given system of equations are of the form (x, y). What is a possible value of x?

(A) -5

(B) -4

(C) -1

(D) 21

@JohannesKP
Highest scoring service

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
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16 Mark for Review

In the xy-plane, the graph of function f, where $y = f(x)$, has exactly 8 x-intercepts. One of these x-intercepts is $(17, 0)$. The rational function g is defined by $g(x) = \frac{f(x)}{x-17}$. In the xy-plane, how many x-intercepts does the graph of $y = g(x)$ have?

Answer Preview:

@JohannesKP
Highest scoring service

17 Mark for Review

$\frac{(x-7)(x-9)}{x-4} = 0$

What is the sum of the solutions to the given equation?

(A) 4

(B) 9

(C) 16

(D) 20

@JohannesKP
Highest scoring service

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
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$-\frac{1}{3}$	-1/3 -.3333 -0.333	-.33 -0.33

18 Mark for Review

In the xy-plane, line s passes through the point $(0, 0)$ and is parallel to the line represented by the equation $y = 42x + 6$. If line s also passes through the point $(4, d)$, what is the value of d?

Answer Preview:

@JohannesKP
Highest scoring service

19 Mark for Review

A circle has center Q, and points A and B lie on the circle. The measure of arc AB is 60°, and the length of this arc is 4 inches. What is the circumference, in inches, of the circle?

(A) 4

(B) 8

(C) 16

(D) 24

@JohannesKP
Highest scoring service

20 Mark for Review

While the mass of an object is the same everywhere, the weight of an object is not the same on different planets. An object has a weight of 170.00 pounds on Earth and a weight of 186.32 pounds on Neptune. The object's weight on Jupiter is 252.8% of its weight on Earth. If the object's weight on Neptune is $x\%$ of its weight on Jupiter, which of the following is closest to the value of x?

(A) 43.35

(B) 71.73

(C) 277.07

(D) 429.76

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Student-produced response directions

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21 Mark for Review

$$-x - wy = -357$$
$$2x - wy = 51$$

In the given system of equations, w is a constant. In the xy-plane, the graphs of these equations intersect at the point $(q, 17)$, where q is a constant. What is the value of w?

Answer Preview:

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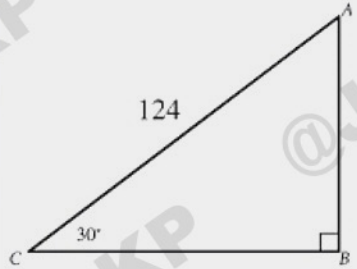
22 Mark for Review



Note: Figure not drawn to scale.

In the right triangle shown, what is the value of $\tan A$?

Answer Preview:

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1 Mark for Review

The function g is defined by $g(x) = 18 - x$. What is the value of $g(3)$?

(A) 3

(B) 15

(C) 18

(D) 21

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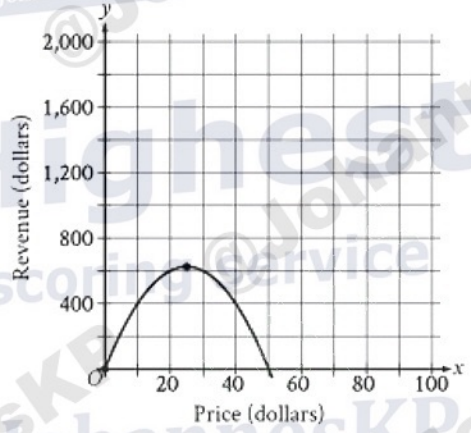
2 Mark for Review



The graph gives the estimated revenue, in dollars, for a company when the price of the company's product is x dollars. What does the y-value of the vertex of the graph represent in this context?

(A) The price, in dollars, of the company's product

(B) The minimum estimated revenue, in dollars

(C) The maximum estimated revenue, in dollars

(D) The estimated revenue, in dollars, when the price of the company's product is 0 dollars

3 Mark for Review

x	y
0	95
1	105
2	115

For a linear relationship between x and y, the table gives three values of x and their corresponding values of y. Which equation represents this relationship?

(A) $y = 10x + 95$

(B) $y = x + 105$

(C) $y = x + 10$

(D) $y = 10x$

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4 Mark for Review

A circle has a radius of 19 meters. What is the area, in square meters, of the circle?

(A) $\frac{19\pi}{2}$

(B) 19π

(C) 38π

(D) 361π

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5 Mark for Review

$p = \sqrt{12c - 25}$

The given equation relates the variables p and c, where c is positive. Which equation correctly expresses c in terms of p?

(A) $c = \frac{(p+25)^2}{12}$

(B) $c = \frac{p^2}{12} + \frac{5}{12}$

(C) $c = \frac{p}{24} + \frac{25}{24}$

(D) $c = \frac{\sqrt{p+25}}{12}$

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6 Mark for Review

The function f is defined by $f(x) = 98x$. What is the slope of the graph of $y = f(x)$ in the xy-plane?

(A) -98

(B) $\frac{1}{98}$

(C) 1

(D) 98

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Student-produced response directions

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7 Mark for Review

A linear model estimates the population of a city from 1993 to 2017. The model estimates the population was 52 thousand in 1993, 219 thousand in 2013, and x thousand in 2017. To the nearest whole number, what is the value of x?

Answer Preview:

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8 Mark for Review

$$w^2 + 8w - 28 = 0$$

Which of the following is a solution to the given equation?

(A) $4 - 2\sqrt{11}$

(B) $2\sqrt{11}$

(C) $\sqrt{11}$

(D) $-4 + 2\sqrt{11}$

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Student-produced response directions

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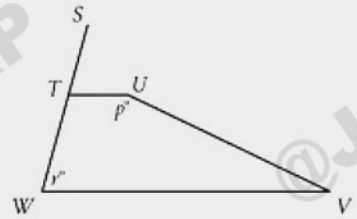
9 Mark for Review



Note: Figure not drawn to scale.

In the figure shown, point T lies on segment SW, and segment TU is parallel to segment VW. The measure of angle V is 26°, and the measure of angle STU is 74°. What is the value of $p - r$?

Answer Preview:

10 Mark for Review

On January 1, there were 233 views of a video posted on a social media site. Every 2 days after January 1, the number of views of the video had increased by 60% of the number of views 2 days earlier. The function f gives the predicted number of views x days after January 1. Which equation defines f?

(A) $f(x) = 233(0.60)^{\frac{x}{2}}$

(B) $f(x) = 233(0.60)^{2x}$

(C) $f(x) = 233(1.60)^{\frac{x}{2}}$

(D) $f(x) = 233(1.60)^{2x}$

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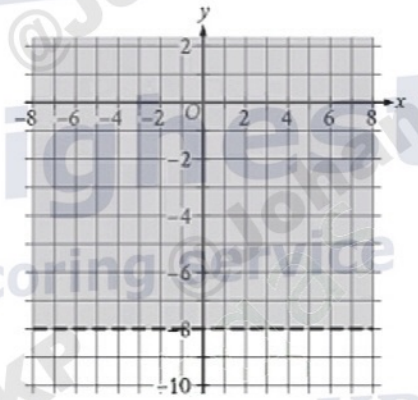
11 Mark for Review



The shaded region shown represents the solutions to the inequality $ry < 128$, where r is a constant. What is the value of r?

(A) 16

(B) 8

(C) −8

(D) −16

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12 Mark for Review

How many solutions does the equation $4(16x - 20) = -16(5 - 4x)$ have?

(A) Exactly one

(B) Exactly two

(C) Infinitely many

(D) Zero

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13 Mark for Review

$$f(x) = 3{,}950(0.38)^{\frac{x}{12}}$$

The function f gives the value, in dollars, of a certain piece of equipment after x months of use. If the value of the equipment decreases each year by $p\%$ of its value the preceding year, what is the value of p?

(A) 3

(B) 8

(C) 38

(D) 62

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14 Mark for Review

A beekeeper's initial observation of the population of a certain bee colony was $1{,}100$ bees. The beekeeper set a goal to increase the population to $2{,}600$ bees. The beekeeper uses a model that predicts the population of this bee colony begins at $1{,}100$ and increases by 120 bees per week in the first two weeks after the initial observation, and then increases by 180 bees per week until the beekeeper's goal is reached. According to this model, at the end of week w after the initial observation, where $w > 2$, which of the following functions gives the predicted number of bees still needed to reach the beekeeper's goal?

(A) $p(w) = 2{,}600 - 180w$

(B) $p(w) = 2{,}480 + 180w$

(C) $p(w) = 1{,}620 - 180w$

(D) $p(w) = -120 + 180w$

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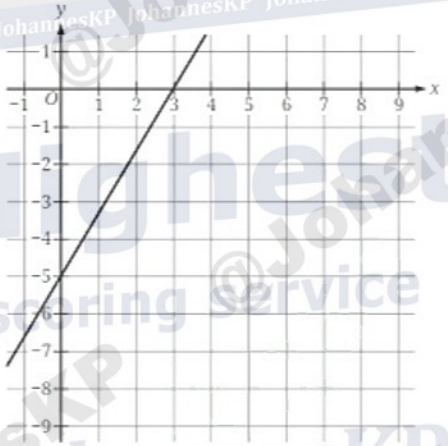
15 Mark for Review



What is an equation of the graph shown?

(A) $5x - 3y = 15$

(B) $5x - 3y = -15$

(C) $3x - 5y = 15$

(D) $3x - 5y = -15$

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16 Mark for Review

Which expression is a factor of $36x^2(x+12)-36(x+12)^3$?

(A) $-12x+144$

(B) $2x+12$

(C) $6x+6$

(D) $24x-144$

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17 Mark for Review

$30x = 1{,}800y - 3{,}600$

One of the two equations in a system of linear equations is given. The system has no solution. Which equation could be the second equation in this system?

A $\frac{1}{30}x = 2y$

B $\frac{1}{30}x = 60y - 120$

C $x = 2y$

D $x = 60y - 120$

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18 Mark for Review

The exponential function f is defined by $f(x) = ab^x$, where a and b are positive constants. If $f(s) = t$ and $f(s+1) = t - 0.87t$, where s and t are constants, what is the value of b?

(A) 0.13

(B) 0.87

(C) 1.13

(D) 1.87

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19 Mark for Review

A container has a volume of 0.84 cubic meters. Which of the following is closest to this volume, in quarts?
(Note: 100.00 centimeters = 1 meter. Use 946.35 cubic centimeters = 1.00 quart.)

(A) 0.09

(B) 7.95

(C) 794.93

(D) 887.62

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20 Mark for Review

Function f is a quadratic function. The graph of $y = f(x)$ in the xy-plane has a vertex at $(4, -6)$, contains the point $(3, -8)$, and has a y-intercept at $(0, a)$. The graph of $y = 5 \cdot f(x)$ has a y-intercept at $(0, b)$. What is the positive difference between a and b?

Answer Preview:

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21 Mark for Review

$$-x^2 + bx - 64 = 0$$

In the given equation, b is an integer. The equation has no real solutions. What is the least possible value of b?

Answer Preview:

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22 Mark for Review

A researcher investigated two species of mites: a predator and its prey. At the start of a week, there was an equal number of the two species. At the end of the week, the number of prey had increased by 2,700% of the number of prey at the start of the week, and the number of predators had increased by 180% of the number of predators at the start of the week. The number of prey at the end of the week was p% greater than the number of predators at the end of the week. What is the value of p?

Answer Preview:

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